Tutorial 05: Fog in the Deferred Renderer

> **Status:** Planned - this tutorial is not written yet.

This tutorial will add **distance and height fog** to the engine as a real, reusable feature - and use it to teach how the deferred pipeline actually works.

Planned topics: - What the G-buffer stores and why the engine reconstructs world position from depth (`inverseViewProjectionMatrix` in the Frame uniforms, 32-byte MRT budget) - How the composition pass consumes per-pixel deferred data - Extending `EnvironmentUniforms` (Scene group) with fog parameters: color, density, height falloff - Implementing the fog blend in `Composition_Deferred.wgsl`, including the horizon / skybox interaction at far depth - Plumbing the parameters CPU-side so fog is configurable per scene

Until then: - Tutorial 04 teaches the render-pass side of the engine: 04_postprocessing.md - The deferred pass order lives in `Renderer::buildPerCameraGraph()` (Shadow -> GBuffer -> ClusterCompute -> Composition -> Skybox -> ForwardTransparency -> Debug)